

5.32%

Defect Rate...

1.9M

Total Output

101K

Total Defect

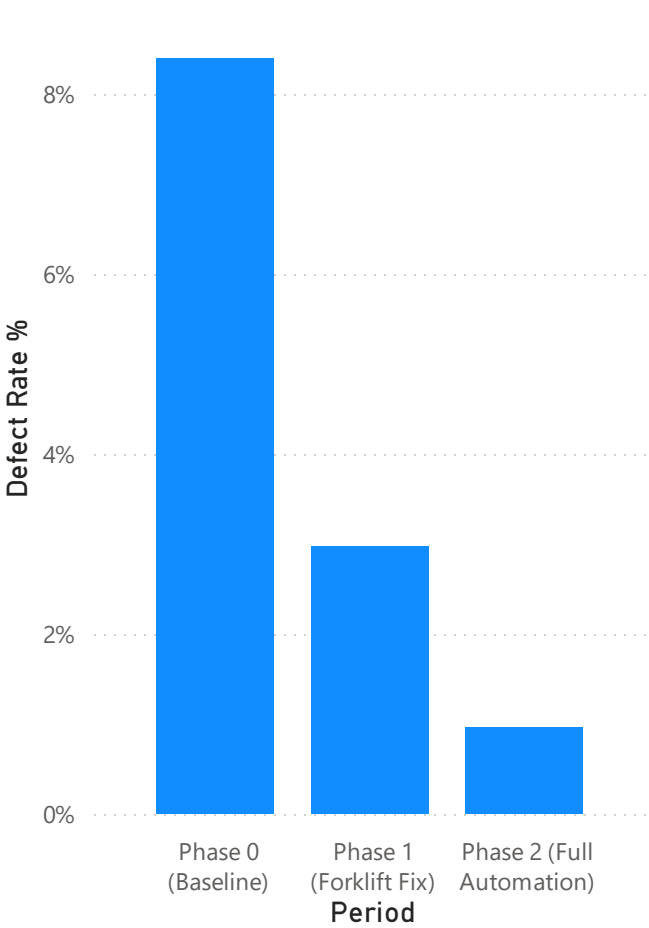
RM761K

Total Rework Cost

6.10%

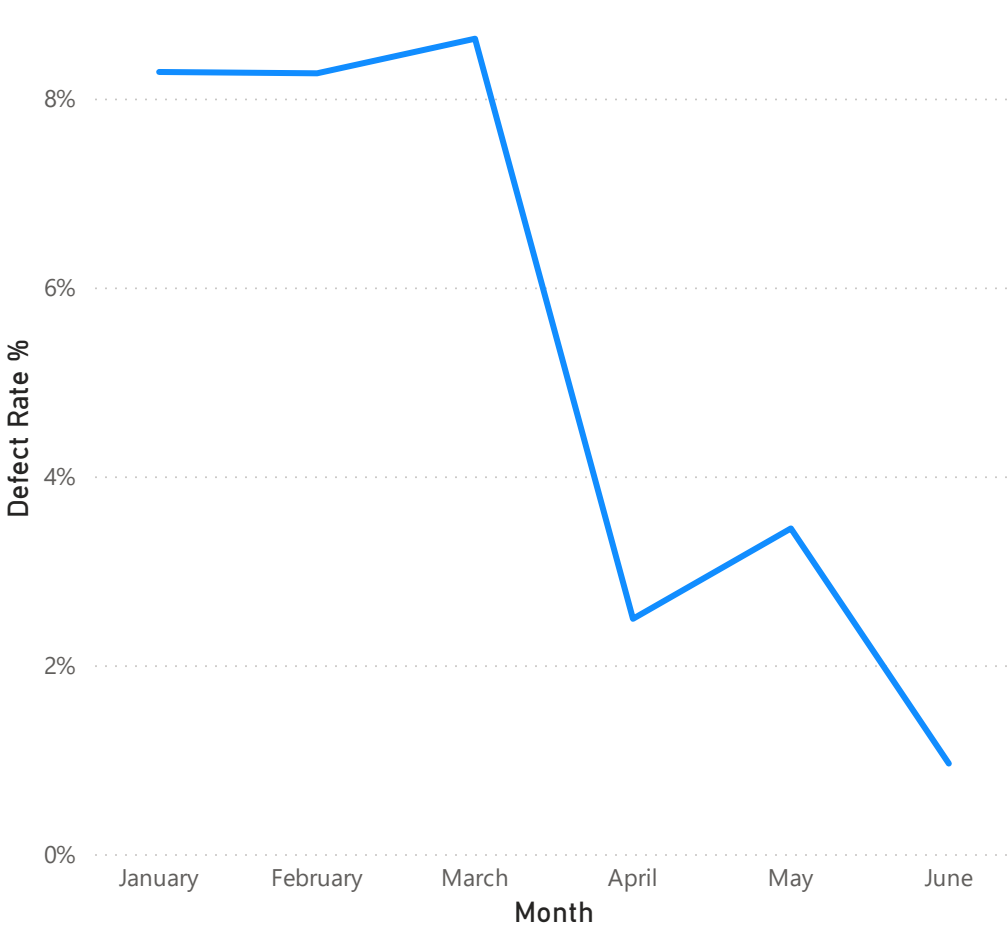
Improvement %

Defect Rate % by Period



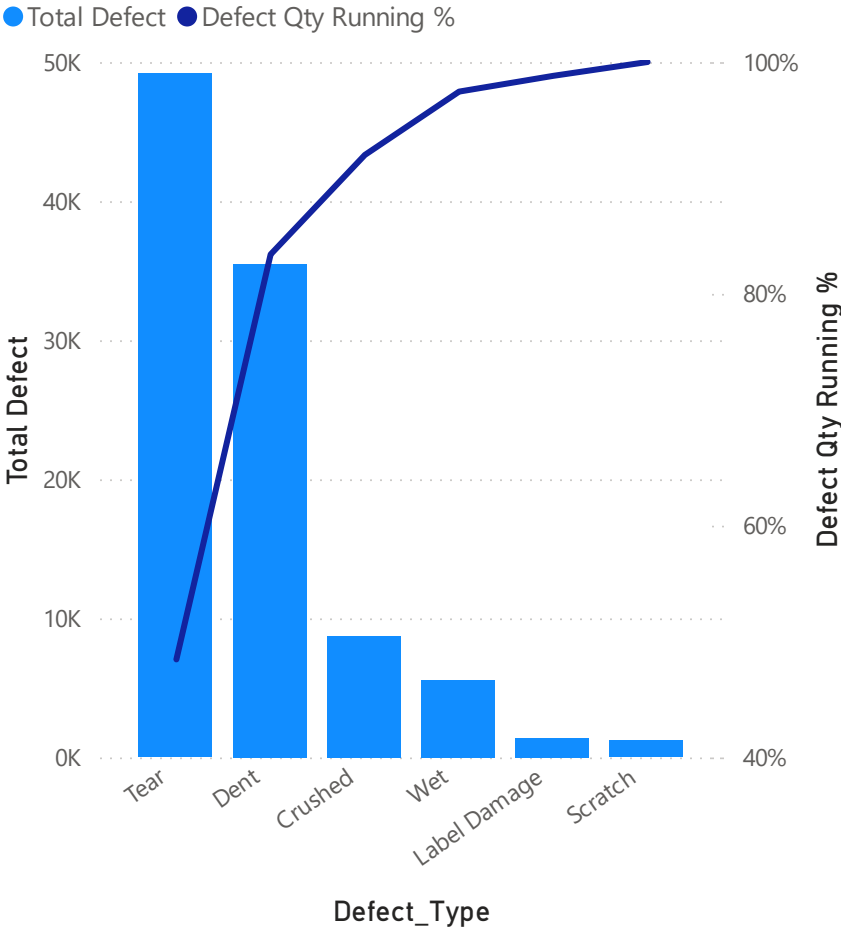
A phase rollout cut the defect rate from 8.4% to 3.0% after fixing forklift handling, then to 1.0% after full automation - an 88% total reduction.

Defect Rate % by Month

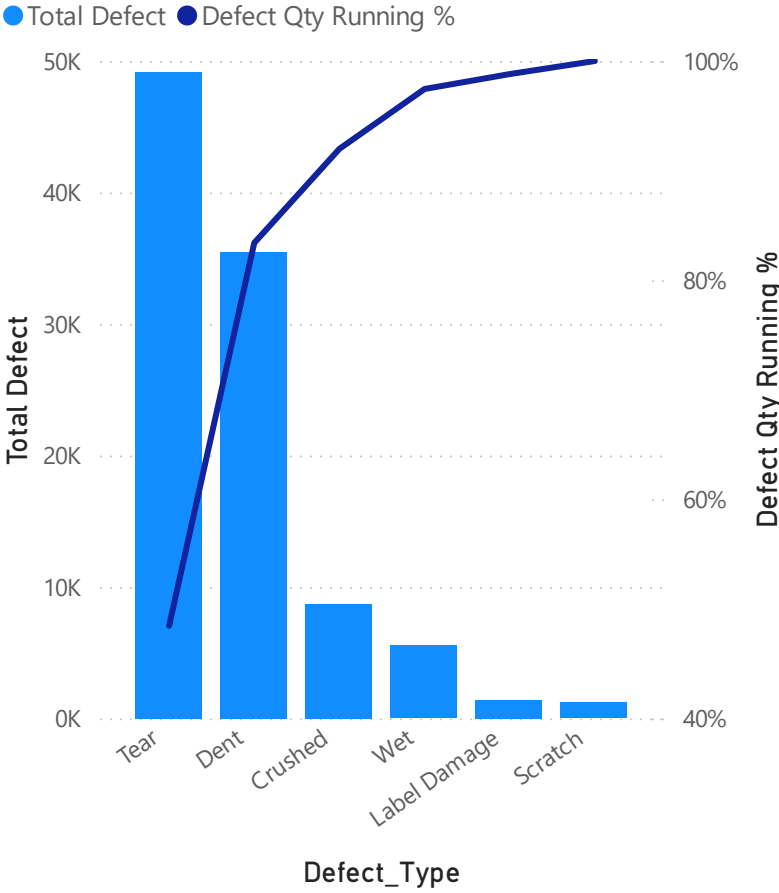


Tear and Dent defects account for 84% of total damage. Forklift-related process changes cut the defect rate from 8.4% to 3.0%, an improvement of 5.4 percentage points.

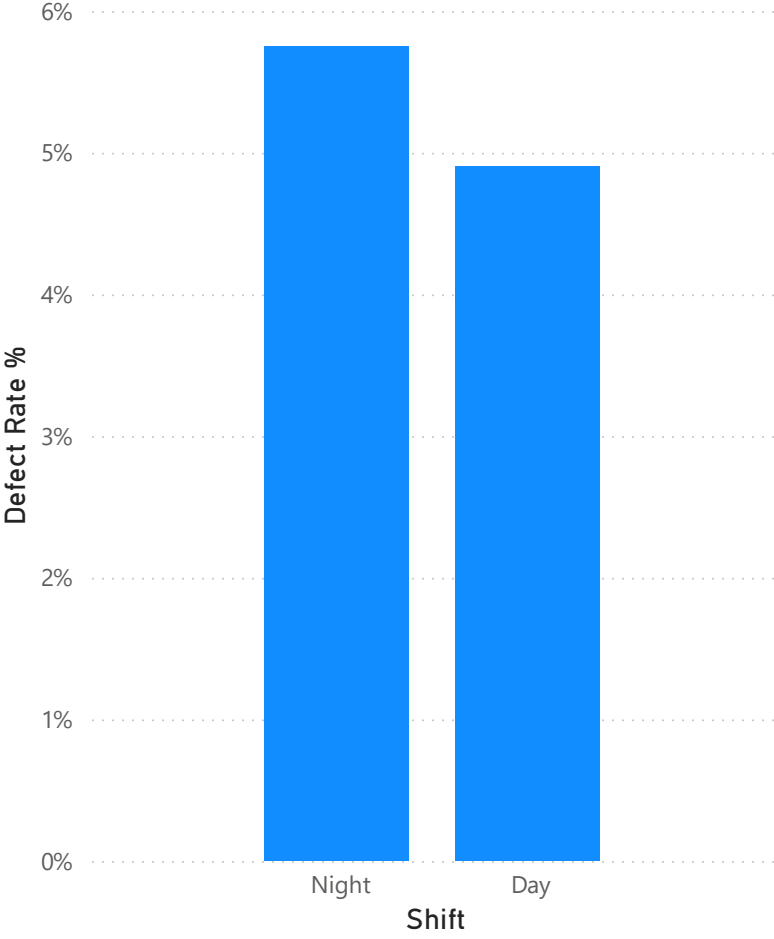
Total Defect and Defect Qty Running % by Defect\_Type



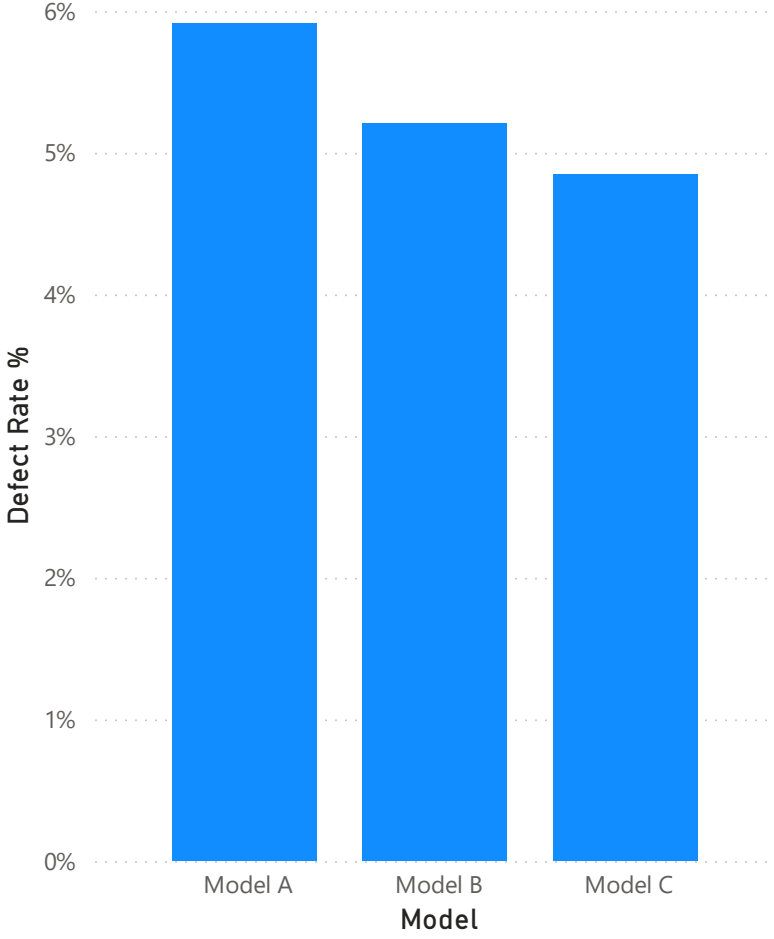
Total Defect and Defect Qty Running % by Defect\_Type



Defect Rate % by Shift



Defect Rate % by Model



Line

Line A

Line B

Line C

Model

Model A

Model B

Model C

Shift

Day

Night

Year and Quarter

☒ 2026

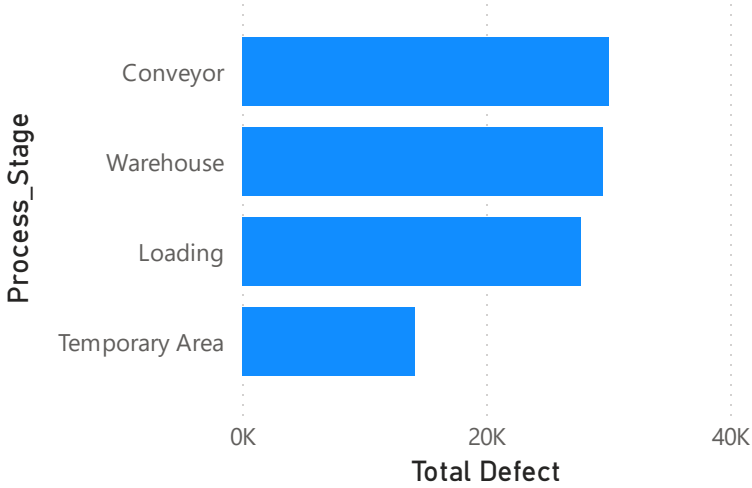
☐ Qtr 1

☐ Qtr 2

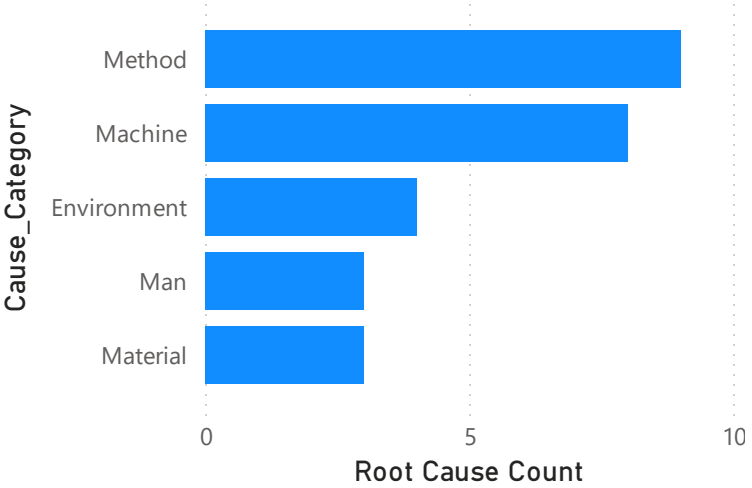
☐ Qtr 3

☐ Qtr 4

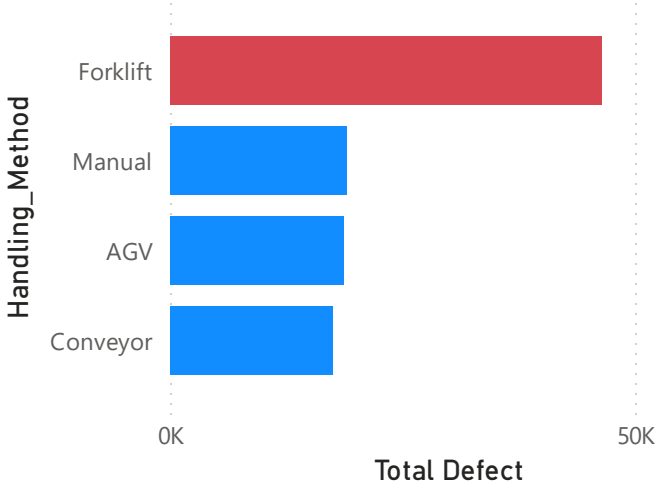
Total Defect by Process\_Stage



Root Cause Count by Cause\_Category



Total Defect by Handling\_Method

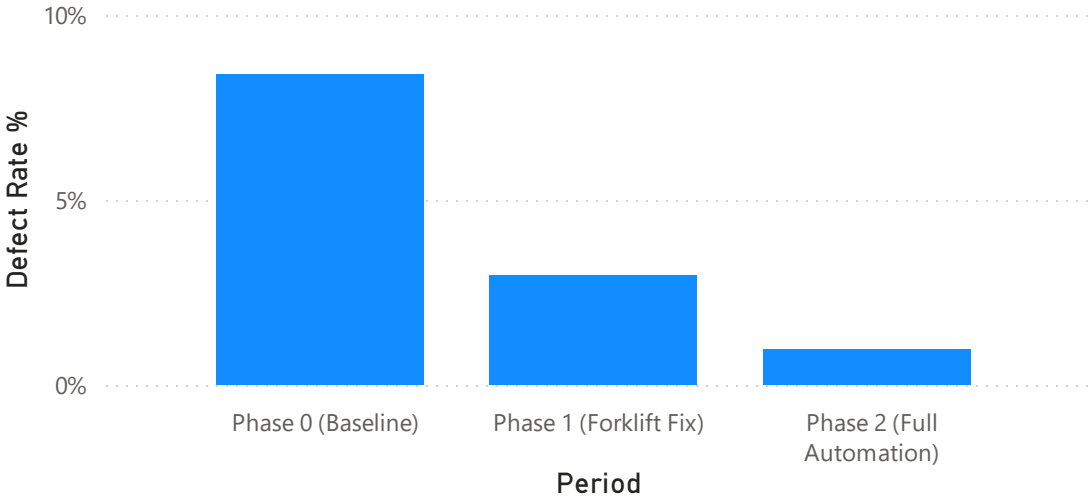


5-Whys

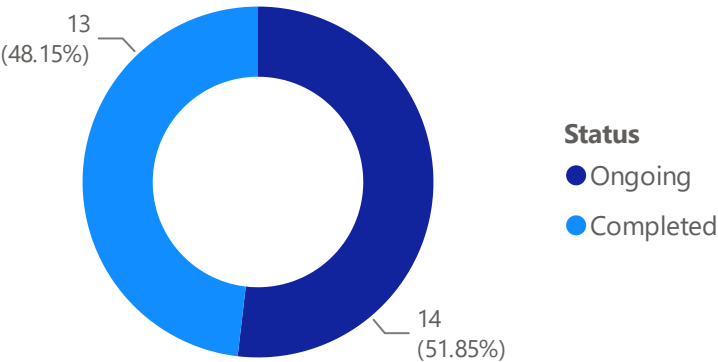
| Why                           | Because                                                                       |
|-------------------------------|-------------------------------------------------------------------------------|
| Why during transfer?          | Handled multiple times using forklift clamp                                   |
| Why was re-handling required? | Production-to-loading flow was not direct                                     |
| Why was the box damaged?      | Torn/dented during transfer                                                   |
| Why was forklioft clamp use?  | Warehouse process required temporary storage for barcode scan and re-handling |

| Process_Stage  | AGV   | Conveyor | Forklift | Manual | Total  |
|----------------|-------|----------|----------|--------|--------|
| Conveyor       | 10814 | 12500    | 5002     | 1742   | 30058  |
| Loading        | 4688  | 3289     | 15644    | 4159   | 27780  |
| Temporary Area | 1847  | 580      | 3813     | 7861   | 14101  |
| Warehouse      | 1301  | 1121     | 21964    | 5174   | 29560  |
| Total          | 18650 | 17490    | 46423    | 18936  | 101499 |

Defect Rate % by Period



Root Cause Count by Status



RM121K  
Total Saving

| Action_ID | Action                                                                                                                       | Owner               | Average of Before_Defect_Rate | Average of After_Defect_Rate | Sum of Saving_MYR |
|-----------|------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------|------------------------------|-------------------|
| A009      | Phase 2: eliminate forklift and warehouse storage; auto-scan labels; flexible motored conveyor loads directly into container | Process Engineering | 3.40                          | 1.00                         | 57773             |
| A001      | Replace forklift transfer with conveyor/AGV flow                                                                             | Process Engineering | 8.50                          | 3.20                         | 24500             |
| A002      | Install pressure-limiting valve on forklift clamps                                                                           | Process Engineering | 8.50                          | 6.10                         | 9800              |
| A006      | Add covered staging area before container loading                                                                            | Facilities          | 4.80                          | 3.90                         | 7600              |
| A004      | Set and enforce max pallet stack height                                                                                      | Warehouse Team      | 7.40                          | 5.90                         | 6100              |
| A005      | Recalibrate pallet wrapping machine tension                                                                                  | Process Engineering | 6.10                          | 4.80                         | 5300              |
| A003      | Retrain forklift and manual handling operators on SOP                                                                        | Warehouse Team      | 8.50                          | 7.40                         | 4200              |
| A007      | Introduce corner protectors for manual stacking                                                                              | Warehouse Team      | 3.90                          | 3.40                         | 3100              |
| A008      | Switch to higher-tack label adhesive                                                                                         | QA                  | 3.40                          | 3.00                         | 2200              |
| Total     |                                                                                                                              |                     | 6.06                          | 4.30                         | 120573            |

| Control Item                         | Frequency      | Target                      | Owner                 | Reaction Plan                                   |
|--------------------------------------|----------------|-----------------------------|-----------------------|-------------------------------------------------|
| Box damage at loading                | Every shipment | 0 tears/dents on inspection | Warehouse Leader      | Stop and inspect handling method                |
| Defect rate by line                  | Daily          | <= 4%                       | Production Supervisor | Investigate if above target                     |
| Forklift contact at Warehous/Loading | Weekly audit   | <= 20% of handling events   | Process Engineer      | Audit clamp setting; retrain operator           |
| Rework cost per 1,000 units          | Monthly        | <= RM 250                   | QA / Process          | Review with management; escalate if trending up |